THE THREE REGIMES OF ESCAPE PATHS FOR ISOSCELES TRIANGLES IN BELLMAN'S LOST-IN-A-FOREST PROBLEM

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Abstract. We organize the best-known escape paths for isosceles triangles $T(\beta)$ (unit legs, base angle β) in Bellman's lost-in-a-forest problem into three regimes, and determine the character of both regime boundaries. On $[45^\circ, 60^\circ]$ the Besicovitch–Movshovich zee is optimal; numerically it remains the frontier down to $\beta_\times = 42.287^\circ$. Below $\beta_\times$ the frontier is a single connected branch of line–arc paths that traverse the boundary of their convex hull minus a chord: near-polygonal trapezoidal *staples* at the top, progressively rounded by arcs whose curvature radius is $\sin\beta$ (the width of the triangle), terminating on the scaled Zalgaller caliper at $\beta^* \approx 27.6^\circ$ — the "Zalgalloid" family Ward speculated about in 2008. The boundaries differ in kind. At $\beta_\times$ two combinatorially distinct branches cross transversally (slopes -0.016 and $+0.013$ per degree): each persists as a local optimum beyond the crossing, the zee resists rounding, and the upper-bound envelope has a corner there — its maximum, so the hardest isosceles forest in the surveyed range sits at the phase boundary, with $\ell(T) \le L_\times = 1.38920$. At β^* nothing crosses: the branch merges tangentially into the caliper, bar length and caliper gap vanishing together. Inside the middle regime we prove two anchors. An explicit piecewise-polynomial family of polygonal staples $S(u)$, $u = \tan(\beta/2)$, escapes and is strictly shorter than every escaping square path for $\beta \in [31.08^\circ, 42.16^\circ]$, and shorter than the diameter, the scaled caliper, and the zee on $[31.50^\circ, 42.16^\circ]$ — so Ward's square path is removed from the frontier everywhere, and the trapezoidal improvement he reported numerically in 2008 becomes a theorem. The proofs eliminate the continuum of positions and headings exactly: escape from a triangle reduces, via the planar case of a containment theorem of Lutwak, to the containment of a disk in an explicit Minkowski sum, and every resulting inequality is a rational-polynomial sign condition verified by exact Sturm certificates. At the type example, the golden gnomon ($\beta = 36^\circ$), an exact line–arc certificate gives $\ell(T) \le 1.2826799$, and the critical line–arc path is characterized algebraically: window tangency forces the curvature radius $\sin\beta$, the sines of its arc angles are roots of explicit quartics over $\mathbb{Q}(\sqrt{5})$, and the critical length $1.2826760\ldots$ has the same closed shape as Zalgaller's strip constant. The branch ends above in an exact rectangle path of length $\sqrt{2}$ at $\beta = 45^\circ$, tied with the diameter. The closed-form bracket $\frac{3}{10}\sqrt{25 - 5\sqrt{5}} \le \ell(T)$ stands at the gnomon. Optimality is claimed nowhere below 45°.

1. Introduction

Bellman [2] asked for the shortest walk guaranteed to leave a forest of known shape when the hiker knows neither their position nor their heading. Formally, a rectifiable path $P \subset \mathbb{R}^2$ is an *escape path* for a compact convex region T if no congruent copy of P lies in $\operatorname{int} T$; the *escape length* $\ell(T)$ is the infimum of the lengths of such paths [9].

Seventy years of improvements. Progress has come one shape — and one mechanism — at a time. Gross [13] settled the disk: the diameter walk is optimal. Isbell [14], completed by Joris [15], settled the half-plane at known distance. Zalgaller [21] settled the infinite strip with the *caliper*, of length $\zeta = 2.2782916\ldots$ per unit width (see also [1, 4]). Gerriets and Poole [10]

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proved that all *fat* regions — those containing a $60°$ rhombus on a diameter — are escaped optimally by a diameter segment. Besicovitch's zigzag [3] for the equilateral triangle was proven optimal by Coulton and Movshovich [5], and Movshovich [17] extended the optimality of the associated *zee* zigzags to all isosceles triangles with base angles in $[45°, 60°]$ (see also [18]).

Why triangles. The solved cases sort into two one-dimensional mechanisms. Fat regions are governed by the *diameter*: the longest chord already escapes optimally. Lean regions are governed by *width*: an arc of width w cannot be hidden in a region of width w, so the scaled caliper gives $\ell(K) \le \zeta\, w(K)$, exact for the strip and asymptotically right for long thin regions. Triangles belong to neither regime: a triangle is never fat (its diameter is one of its sides, and the rhombus needs room on both sides of that chord), and it is lean only in the limit of small base angle. They are thus the simplest forests squeezed strictly between the two solved mechanisms, and each solved triangular case has required a genuinely new path: the zigzags of Besicovitch and Movshovich on $[45°, 60°]$. Below $45°$ the problem is open, and the isosceles family $T(\beta)$ — unit legs, base angles β, base $2\cos\beta$ — is its standard laboratory.

Ward [19] explored this open range numerically. He compared the scaled caliper, the zee, the diameter, and a fourth candidate, the *square path* (three sides of a square), and found the square path shortest among the four for $32.36° < \beta < 41.34°$; from this he concluded (his Proposition 15) that "there exists at least one more type of solution for bounded forests." He then recorded the following, which we quote in full since it is the subject of this paper:

> "Some numerical experiments indicate that the results for the square can be improved slightly by increasing the lower base to form a path with an isosceles trapezoidal convex hull. This leads to shorter path lengths and a slightly larger range over which the path surpasses Zalgaller and Besicovitch." [19, p. 49]

He published no coordinates or lengths for this improvement. Gibbs [11], using an uncertified Monte-Carlo search, later placed a "trapezium" phase near base angles $38°$–$42°$ and line–arc "tunnel" candidates below it; in a companion paper [12] he argued that for convex polygonal forests some shortest escape path is piecewise linear–circular. We return to Ward's "Zalgalloid" question in Sections 6 and 7, with an affirmative (numerical) answer. Recent work of Deng [6, 7] formulates the general problem as a boundary-intersection problem and studies finite traveling-salesman discretizations. Our aim is complementary: for triangular forests the continuum of positions and headings can be eliminated *exactly* — a path escapes if and only if one explicit Minkowski sum contains a disk (Section 2) — and candidate paths then admit finite, checkable certificates. No sampled positions or orientations occur in any proof below.

The three regimes. This paper organizes everything that is known about escape paths for the family $T(\beta)$ — proven or numerical — into a single picture, Figure 1. Three families take turns as the shortest known escape path; one representative of each, at its critical placement, is drawn in Figure 2.

- The **zee** of Besicovitch and Movshovich, for $\beta \ge \beta_\times = 42.287°$: a three-segment zigzag that visits the four vertices of its quadrilateral hull in *diagonal* order, so its middle segment crosses the hull. On $[45°, 60°]$ it is proven optimal [5, 17]; on $[\beta_\times, 45°)$ it remains the shortest known path (Section 3).
- The **staple–Zalgalloid branch**, for $\beta^* \approx 27.6° \le \beta \le \beta_\times$: a single connected family of *boundary-type* paths, each traversing the boundary of its convex hull minus one chord. Near $\beta_\times$ the hull is a near-polygonal isosceles trapezoid — the *staple*; as β decreases, circular arcs of curvature radius $\sin\beta$, the width of $T(\beta)$, grow along it (Sections 4–6).
- The **scaled caliper**, for $\beta \le \beta^*$: Zalgaller's strip solution scaled to the width of the triangle, $\ell(T) \le \zeta \sin\beta$ (Section 7).

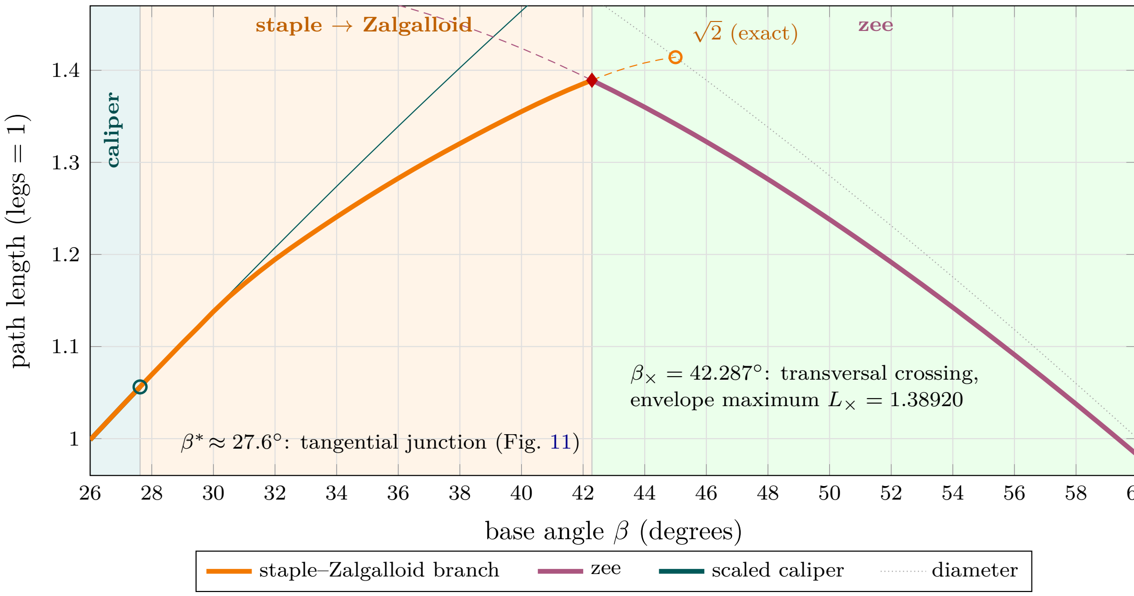

FIGURE 1. The three regimes of best-known escape paths (legs = 1): the scaled caliper for $\beta \le \beta^* \approx 27.6°$, the staple-to-Zalgalloid line–arc branch on $[\beta^*, \beta_\times]$, and the zee beyond $\beta_\times = 42.287°$. Thick curves are the frontier; dashed curves continue each branch past its regime, where it persists as a local optimum. The crossing at $\beta_\times$ is transversal, and the kink is the *maximum* of the envelope: the hardest isosceles forest sits at the regime boundary (diamond). At β^* the branch instead merges tangentially into the caliper (circle). The branch terminates above in an exact rectangle path of length $\sqrt{2}$ at $\beta = 45°$ (Proposition 6.3), tied with the diameter. Regime boundaries and dashed data are numerical (Sections 3.1 and 7); the zee is proven optimal on $[45°, 60°]$ [5, 17]; Theorems 4.1 and 6.1 certify anchors inside the middle regime.

Two boundaries, two characters. The organizing observation of the paper is that the two regime boundaries could hardly be more different.

At $\beta_\times$ two combinatorially distinct families cross *transversally*. The length curves meet with slopes -0.016 (zee) and $+0.013$ (branch) per degree; each family continues past the crossing as a strict local optimum — the branch up to an exact rectangle path of length $\sqrt{2}$ at $\beta = 45°$ (Proposition 6.3), the zee at least down to 36° — and no continuous deformation connects them: freeing a corner-rounding radius in the zee returns zero at every tested angle. The envelope of best known lengths therefore has a corner at $\beta_\times$ — a *first-order* phase boundary — and the corner points up: it is the *maximum* of the envelope, so the hardest isosceles forest with unit legs in the surveyed range is the one at the phase boundary, with $\ell(T(\beta_\times)) \le L_\times = 1.38920$ (Section 3.1).

At β^*, by contrast, nothing crosses. The branch itself deforms onto the caliper: its bar length and its margin over the caliper vanish together, tangentially, leaving no kink in the envelope — a *second-order* boundary (Section 7).

Certified results, and what is numerical. The regime picture is numerical where it must be — the boundary locations, the slopes, the identity of the frontier family — and exact where we can make it so. Four anchors are theorems.

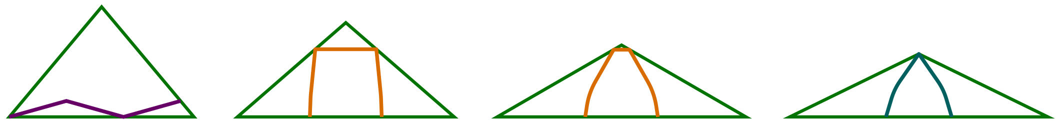

Figure 2. One escape path per regime, each at a critical placement in its triangle $T(\beta)$ (legs = 1; the hull fits only in touching position). Left to right: the *zee* at $\beta = 50°$ — diagonal type, the middle segment crosses the hull; the *smoothed staple* at $\beta = 41°$ — boundary type: bar under the apex, shoulder arcs of radius $\sin\beta$ and span $3.5°$, to the eye still a trapezoid; a *Zalgalloid* at $\beta = 30°$ — the same branch with deep arcs and a short bar; and the *scaled caliper* at $\beta = 26°$, which the branch has joined (bar length zero). Lengths are the best known values at each angle; placements are computed, not sketched.

- *The interval theorem* (Theorem 4.1): an explicit piecewise-polynomial family of polygonal staples $S(u)$, $u = \tan(\beta/2)$, escapes $T(\beta)$ and is strictly shorter than every escaping square path on $[31.08°, 42.16°]$, and shorter than Ward's full quartet — diameter, caliper, zee, squares — on $[31.50°, 42.16°]$. This removes the square path from the frontier everywhere and turns the trapezoidal improvement Ward reported in 2008 into a theorem, with the enlarged range he predicted certified at both ends.
- *The gnomon certificates* (Theorems 5.1 and 6.1): at the type example $\beta = 36°$, the golden gnomon, the closed-form polygonal bound 1.2849616, improved by the first exact arc-window certificate in this problem area to $\ell(T) \leq 1.2826799$.
- *The critical smoothed staple* (Proposition 6.2): window tangency forces arcs of curvature radius $\sin\beta$, the width of T; at the gnomon the sines of the critical arc angles are roots of explicit quartics over $\mathbb{Q}(\sqrt{5})$, and the critical length $1.2826760\ldots$ has the same closed shape as Zalgaller's strip constant.
- *The rectangle endpoint* (Proposition 6.3): the branch terminates at $\beta = 45°$ in an exact rectangle path of length $\sqrt{2}$, tied with the diameter of the right isosceles triangle.

The engine behind all four is the same finite certificate (Section 2): escape from a triangle reduces, via the planar case of a containment theorem of Lutwak, to the containment of a disk in one explicit Minkowski sum, and each containment becomes a finite list of exact sign conditions — Sturm certificates over an interval of angles, algebraic signs at a single triangle. Conversely, every statement obtained by numerical optimization is labeled as such where it appears and enters no proof; the boundaries $\beta_\times$ and β^* are of this kind. We emphasize what is *not* claimed: optimality of any path below $45°$. The sections walk down the β axis: the criterion (Section 2), the zee regime and its first-order end (Section 3), the certified staple interval (Section 4) and its exact type example (Section 5), the rounded staple (Section 6), and the caliper regime with the second-order junction (Section 7). All code and data accompanying the paper are available at `https://github.com/atemerev/bellman-staple` and as arXiv ancillary files.

2. The escape criterion

Throughout, $h_C(u) = \sup_{x\in C}\langle x, u\rangle$ denotes the support function of a compact convex set C, and R_θ the rotation of $\mathbb{R}^2$ by angle θ. We use two standard facts: $h_{C\oplus D} = h_C + h_D$ for the Minkowski sum, and $h_{RC}(u) = h_C(R^{-1}u)$.

Lemma 2.1 (fit criterion for triangles). *Let $T = \{x : \langle n_i, x\rangle \le t_i,\ i = 1,2,3\}$ be a triangle with unit outward normals n_i and side lengths s_i, so that $\sum_i s_i n_i = 0$ and $\sum_i s_i t_i = 2\,\mathrm{Area}(T)$. For a compact convex C there exists a translation t with $C + t \subseteq T$ if and only if*

$$s_1 h_C(n_1) + s_2 h_C(n_2) + s_3 h_C(n_3) \ \le\ 2\,\mathrm{Area}(T).$$

Since $\sum_i s_i\, h_C(n_i) = 2V(C,T)$, twice the mixed area, the criterion reads: *C translates into T if and only if $V(C,T) \le V(T,T) = \mathrm{Area}(T)$.* In this mixed-volume form (for simplices in all dimensions) the criterion is a result of Lutwak [16, Prop., p. 230 and the remark following Cor. 3], with ancestry in Weil's characterization of translative containment [20]. We include the short linear-programming proof both to keep the note self-contained and because its sufficiency half produces the explicit feasible translations used in our certificates. We have not found this finite disk-containment specialization in the escape-path literature.

Proof. $C + t \subseteq T$ is equivalent to the linear system $\langle n_i, t\rangle \le c_i := t_i - h_C(n_i)$, $i = 1,2,3$.

Necessity. If t solves the system then $0 = \langle \sum_i s_i n_i, t\rangle \le \sum_i s_i c_i$, which rearranges to the stated inequality.

Sufficiency. Suppose $\lambda := \big(\sum_i s_i c_i\big)/\big(2\,\mathrm{Area}(T)\big) \ge 0$. The affine system $\langle n_i, t_0\rangle = c_i - \lambda t_i$ ($i = 1,2,3$; three equations in two unknowns) is solvable: the unique linear dependency among the rows n_i has coefficients s_i, and $\sum_i s_i(c_i - \lambda t_i) = \sum_i s_i c_i - \lambda \cdot 2\,\mathrm{Area}(T) = 0$ is precisely its compatibility condition. For this t_0,

$$\{t : \langle n_i, t\rangle \le c_i\} \ =\ \lambda T + t_0 \ \ne\ \emptyset. \qquad \square$$

Note the geometric content of sufficiency: *the set of admissible translations is itself a homothet of the triangle*, of ratio λ.

Proposition 2.2 (escape criterion). *With T, $n_i = (\cos\alpha_i, \sin\alpha_i)$, s_i as above, a path P with $H = \mathrm{conv}(P)$ is an escape path for T if and only if the Minkowski sum*

$$W \ :=\ s_1\, R_{-\alpha_1} H \ \oplus\ s_2\, R_{-\alpha_2} H \ \oplus\ s_3\, R_{-\alpha_3} H$$

contains the closed disk of radius $2\,\mathrm{Area}(T)$ centered at the origin. If H is a polygon, then so is W.

Proof. A placed congruent copy of P lies in the open convex set $\mathrm{int}\, T$ if and only if its convex hull does, and rigid motions factor as a rotation followed by a translation; if a placed copy is not contained in the interior, the connected path, starting inside, must meet ∂T — the hiker escapes (cf. [9]). Thus P escapes iff for every θ no translate of $R_\theta H$ lies in $\mathrm{int}\, T$; by Lemma 2.1 this holds iff for all θ, $\sum_i s_i\, h_{R_\theta H}(n_i) \ge 2\,\mathrm{Area}(T)$ (the closed inequality: a hull that can only be translated to touch ∂T still reaches the boundary). Writing $h_{R_\theta H}(n_i) = h_H(R_{-\theta} u(\alpha_i)) = h_{R_{-\alpha_i} H}(u(-\theta))$ with $u(\psi) = (\cos\psi, \sin\psi)$ and summing, $\sum_i s_i h_{R_\theta H}(n_i) = h_W(u(-\theta))$. The condition $h_W(u) \ge 2\,\mathrm{Area}(T)$ for every unit u is equivalent to $W \supseteq \overline{D}(0, 2\,\mathrm{Area}(T))$. $\square$

Remark 2.3. Two computational forms of the inclusion are used below. For a fixed triangle, W is an explicit polygon and the inclusion is a list of edge-distance inequalities (Section 5 and Figure 8). For a one-parameter family it is more convenient to avoid tracking the boundary of W: a convex polygon is the intersection of its edge half-planes, and every edge normal of a Minkowski sum is an edge normal of one of the summands, so it suffices to verify $h_W(\nu) \ge 2\,\mathrm{Area}(T) \cdot |\nu|$ for the twelve candidate directions $\nu = R_{-\alpha_i} g$, with g ranging over the four edge normals of H — and $h_W(\nu)$ may be replaced by any convenient lower bound, e.g. by evaluating each summand's support at a fixed vertex (Section 4).

Remark 2.4. For the unit equilateral triangle the criterion reads: a path escapes iff the Minkowski sum of three copies of its hull rotated by $0°, 120°, 240°$ contains the disk of radius $2\,\mathrm{Area} = \frac{\sqrt{3}}{2}$. Besicovitch's zigzag [3, 5] is exactly critical in this sense: its Minkowski sum circumscribes that disk.

3. The zee regime

For base angles in $[45°, 60°]$ the problem is solved, and the solution is the *zee* z_β: three segments of equal length making angles $+\delta$, $-\delta$, $+\delta$ with a common axis, where

$$|z_\beta| = \frac{6\sin\beta\cos\beta}{\sqrt{1+8\sin^2\beta}}, \qquad \tan\delta = \frac{1}{3\tan\beta}$$

(the z_β of [18, 17]; the closed form is Ward's [19], rescaled to legs 1). At the equilateral corner $\beta = 60°$ this is Besicovitch's zigzag [3], of length $0.981980\ldots$, proven optimal by Coulton and Movshovich [5]; Movshovich [17] extended optimality to the whole range $[45°, 60°]$. Combinatorially the zee is the *diagonal* realization of its convex hull: the four path vertices span a quadrilateral, and the path visits them in the order that sends its middle segment across the interior (Figure 2, first panel).

Below $45°$, two numerical facts about the zee frame the rest of the paper. First, nothing happens to the zee itself: it remains a valid and *exactly critical* escape path at least down to $\beta = 36°$ — critical in the sense that its Minkowski sum (Proposition 2.2) circumscribes the disk — and it is irreducibly polygonal: re-optimizing its combinatorial type with a free corner-rounding radius returns radius zero at every angle in $[36°, 60°]$, consistent on $[45°, 60°]$ with the proven optimality of the polygonal path. Second, the zee is nonetheless overtaken — and abruptly so.

3.1. **The crossing at the end of the zee regime is first-order.** *Everything in this subsection is numerical; it enters no proof.* Optimizing the zee family and the boundary-type branch of Sections 4–6 against the escape criterion, with cutting-plane refinement of the direction constraints, the two length curves cross at

$$\beta_\times = 42.287°, \qquad \ell\bigl(T(\beta_\times)\bigr) \;\le\; L_\times \;=\; 1.38920,$$

in striking agreement with Gibbs's Monte-Carlo estimate $42.3°$ [11]; the purely polygonal crossing — optimal trapezoid against zee, the right-hand dot in Figure 4 — lies at $42.281°$, and Ward's square-path crossover at $41.34°$ [19] is superseded together with the square path itself (Theorem 4.1). Three observations make this boundary *first-order* — a transversal crossing of distinct branches, not a deformation of one branch into the other.

1. *The slopes differ.* Near $\beta_\times$ the zee curve falls at -0.0163 per degree while the branch curve rises at $+0.0132$ per degree: the envelope of best known lengths has a corner with slope gap 0.0294 per degree. And the corner points *up*, so $L_\times$ is the maximum of the envelope over the surveyed range $[26°, 60°]$: the hardest isosceles forest with unit legs sits exactly at the phase boundary.
2. *Both families outlive the crossing.* The boundary-type branch continues above $\beta_\times$ as a strict local optimum all the way to an exact rectangle endpoint at $45°$ (Proposition 6.3); the zee continues below, valid and critical, at least to $36°$.
3. *No family connects them.* The zee realizes its hull in diagonal order and refuses curvature; the branch realizes its hull in boundary order and, below $\approx 42.9°$, demands curvature (Section 6.3).

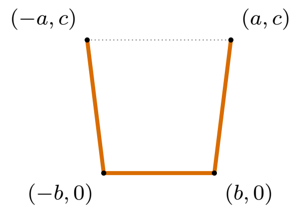

Figure 3. A staple and its isosceles-trapezoidal hull: the bar from $(-b, 0)$ to $(b, 0)$, two equal legs to $(\pm a, c)$, and the dotted top edge completing the hull.

These are statements about the two specified families, not proven phase boundaries of the unrestricted problem; what is proved is the interval of Corollary 4.2.

4. The staple regime: a certified interval

Below $\beta_\times$ the frontier belongs to paths that wrap their hull, and the simplest of these are polygonal. A *staple* is a three-segment path whose four vertices $(\pm b, 0), (\pm a, c)$ form an isosceles trapezoid: a bar of length $2b$ with two equal legs attached at equal obtuse angles (Figure 3). Ward's square path is the case $a = b$, $c = 2b$; releasing the right angles is what saves length. The name is borrowed from Movshovich and Wetzel: their *staples* [18, §3.2], after [5], are the three-segment arcs whose endpoints lie on the base of the triangle and whose two middle vertices lie on the two arms — exactly the contact pattern of our path at a critical placement (Figure 7) — and their Lemma 9 shows that in the zee range, base angles above 45°, every staple is too long to be optimal. Theorem 4.1 certifies that below 45° the staples take the lead.

Theorem 4.1. *Let $u = \tan(\beta/2)$, and let $S(u)$ be the staple whose parameters $a(u), b(u), c(u)$ are the explicit rational polynomials of Section 4 (degrees at most 6; 51 rational coefficients in total, printed in Appendix A and listed in the ancillary file* `interval_construction.json`*). Then for every $u \in [0.278, 0.3855]$, that is for every base angle $\beta = 2 \arctan u \in [31.0719°, 42.1633°]$:*

(i) *$S(u)$ is an escape path for $T(\beta)$;*
(ii) *$|S(u)| < 3s$ for every square path of side s that escapes $T(\beta)$ — in particular $|S(u)|$ is strictly smaller than Ward's square-path length in both of its regimes;*
(iii) *$|S(u)|$ is strictly smaller than the critical zee length $\frac{6 \sin\beta \cos\beta}{\sqrt{1+8\sin^2\beta}}$ (Ward's closed form [19], rescaled to legs 1; the arcs are the z_β of [18, 17]) and the diameter $2\cos\beta$; and for $u \geq 0.282$ (i.e. $\beta \geq 31.4969°$) also strictly smaller than the caliper length $\zeta \sin\beta$.*

Corollary 4.2 (Ward's enlargement, certified)**.** *For every base angle $\beta \in [31.50°, 42.16°] \supsetneq (32.36°, 41.34°)$, the staple $S(u(\beta))$ is an escape path for $T(\beta)$ strictly shorter than each of the four candidates compared by Ward: the diameter, the scaled Zalgaller caliper, the Besicovitch–Movshovich zee, and every escaping square path.*

Figure 4 shows the five length curves. Two features are the content of Theorem 4.1: the staple curve lies strictly below the square-path curve across a band that covers Ward's entire square-path range, so the square path is now nowhere the best known candidate; and it lies below all four competitors on $[31.50°, 42.16°]$, certifying the enlargement Ward predicted — at both ends.

The rest of this section proves Theorem 4.1. Parametrize by $u = \tan(\beta/2)$, so that $\sin\beta = \frac{2u}{1+u^2}$ and $\cos\beta = \frac{1-u^2}{1+u^2}$ are rational functions of u. All triangle data — normals, side lengths, the rotations of Proposition 2.2, the disk radius $2\,\text{Area} = \sin 2\beta$ — are then rational in u, and for a

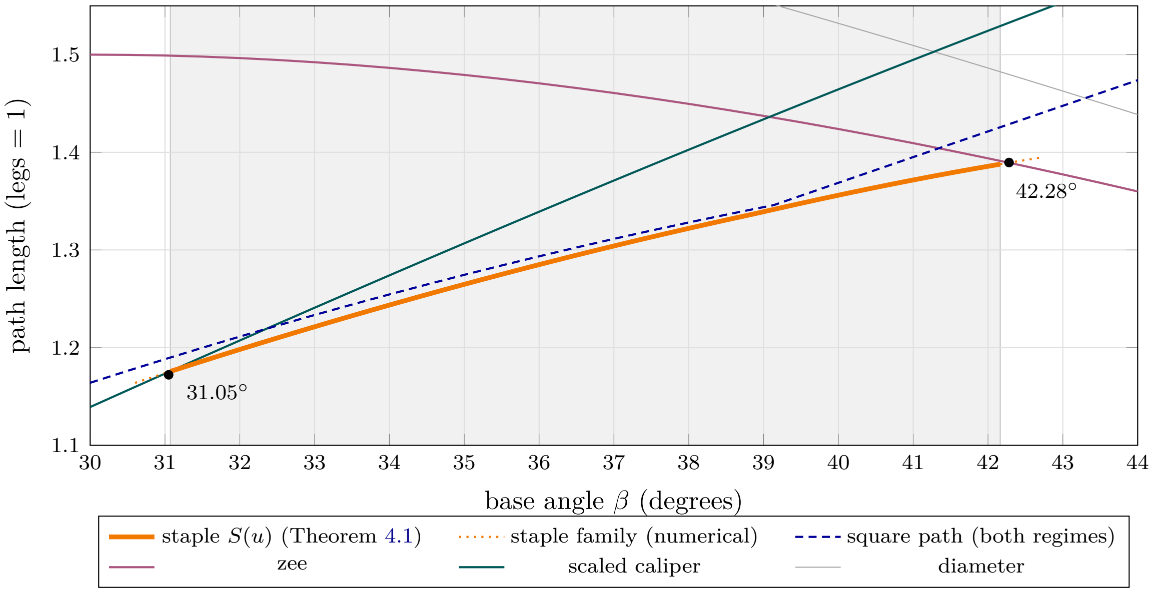

FIGURE 4. The five candidate lengths against the base angle β. The shaded band $[31.07°, 42.16°]$ is the certified domain of Theorem 4.1: there the staple $S(u)$ (thick) escapes and lies strictly below the square-path curve, whose two regimes meet in the visible kink at $\beta_0 = 39.132°$. On $[31.50°, 42.16°]$ it lies below all four competitors (Corollary 4.2). Dotted continuations and the two crossing points at 31.05° and 42.28° are numerical (Sections 3.1 and 7); everything else in the band is proved.

staple whose parameters a, b, c are *polynomials* in u with rational coefficients, every certificate of Remark 2.3, after clearing the (positive) denominators and squaring away the (sign-certified) square roots, is the strict positivity of a polynomial in $\mathbb{Q}[u]$ on a rational interval — decidable by Sturm's theorem in exact integer arithmetic.

4.1. **The construction.** The ancillary file `interval_construction.json` defines three polynomial triples $(a_i, b_i, c_i) \in \mathbb{Q}[u]^3$, of degrees 6, 2, and 6 on the respective pieces

$$[0.278,\ 0.3530], \qquad [0.3530,\ 0.353399], \qquad [0.353399,\ 0.3855],$$

and $S(u)$ is the path through $(-a_i(u), c_i(u))$, $(-b_i(u), 0)$, $(b_i(u), 0)$, $(a_i(u), c_i(u))$. All 51 coefficients are rational numbers, printed in full in Appendix A; the machine-readable copy is read afresh by the verification script, which certifies $a_i, b_i, c_i > 0$ together with every inequality used below. Figure 5 shows the family at three angles. The middle piece is a deliberately thin bridge: near $u \approx 0.3534$ the numerically optimal staple changes which edges of its Minkowski sum touch the disk (Section 4.4), and the two outer fits track the two contact regimes; the quadratic bridge hands over between them, and its exact endpoints are immaterial — every certificate is checked on each piece separately.

4.2. **Eliminating square paths.**

Lemma 4.3 (square paths from inscribed squares)**.** *Fix β and suppose some square Q_0 of side s_0 satisfies $Q_0 \subseteq T(\beta)$. Then every square path of side $s < s_0$ fails to escape $T(\beta)$; consequently every escaping square path has length $\geq 3s_0$.*

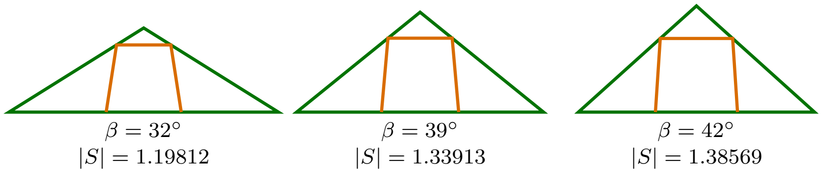

FIGURE 5. The certified family $S(u)$ inside $T(\beta)$ at $\beta = 32°$, $39°$, $42°$, each at a critical placement (the hull fits only in touching position).

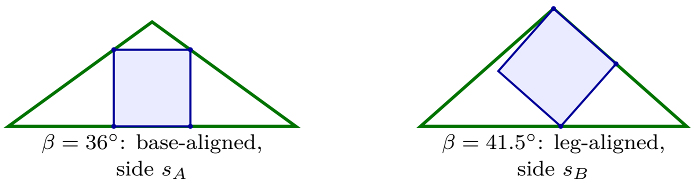

FIGURE 6. The two inscribed squares behind Lemma 4.3. Left: the base-aligned square, with both bottom corners on the base and both top corners on the legs. Right: the leg-aligned square, with one full side on a leg, its far corner at the apex, and one corner on the base (contacts marked). Any escaping square path must have side at least $\max(s_A, s_B)$; the certificates prove $|S(u)| < 3s_A$ for $u \leq 0.365$ and $|S(u)| < 3s_B$ for $u \geq 0.36$, which together cover the whole interval.

Proof. The hull of a square path of side s is the full square. Shrinking Q_0 about its center by the factor $s/s_0 < 1$ places a congruent copy of that hull in $\operatorname{int} Q_0 \subseteq \operatorname{int} T(\beta)$. $\square$

The two inscribed squares we use are Ward's, in exact form (Figure 6): the base-aligned square of side

$$s_A = \frac{2 \sin\beta \cos\beta}{\sin\beta + 2\cos\beta},$$

with both bottom corners on the base and both top corners on the legs, and the leg-aligned square of side

$$s_B = \frac{\sin\beta}{\sin\beta + \cos\beta},$$

with one full side on a leg, its far corner *at the apex*, and one corner on the base. The stated incidences are polynomial identities in u; the remaining vertex–halfplane constraints are certified by Sturm checks. Ward's two square-path formulas $3s_A$ and $3s_B$ exchange at the root $\beta_0 = 39.1320\ldots°$ of $\sin\beta\,(2\cos\beta - 1) = 2\cos\beta\,(1 - \cos\beta)$ (his "approximately 39.1°"), the kink visible in Figure 4.

4.3. **The certificates.** Every assertion of Theorem 4.1 is now a strict polynomial inequality on a rational interval. (i): the twelve support-dominance inequalities of Remark 2.3 per piece, with fixed support-test vertices (their values are lower bounds for h_W); for the four leg-normal directions the inequality is squared once, with h-positivity certified separately. (ii): by Lemma 4.3 it suffices that $|S(u)| < 3s_A(u)$ for $u \leq 0.365$ and $|S(u)| < 3s_B(u)$ for $u \geq 0.36$, once the two squares are certified inscribed; each comparison squares away the single root in $|S(u)|$. (iii): the diameter and caliper comparisons are of the same shape, using the outward-rounded bracket $\zeta > 2.2782916$ from the published closed form; the zee comparison carries one further square root and is squared twice, with both intermediate signs certified. (The zee length in (iii) is Ward's closed form Q_B [19] rescaled from unit base to unit legs.)

In total 116 rational-polynomial certificates (degrees ≤ 40) are verified by endpoint signs and exact Sturm root counts. The Zalgaller constant is the only transcendental input and is handled separately by directed evaluation. The verification script is the ancillary file `interval_certificate.py`; it re-derives every polynomial from the construction data and the exact triangle geometry. No floating point enters any accepted certificate. □

Corollary 4.2 follows by intersecting the intervals in Theorem 4.1(ii)–(iii): all four comparisons hold simultaneously for $u \in [0.282, 0.3855] \supseteq [31.50°, 42.16°]$ in base angle. □

4.4. **How the witnesses were found.** *This subsection is numerical; it enters no proof.* A stationary point within a family proves nothing about optimality, within the family or otherwise; it only tells us where to place a certificate.

For a symmetric staple at the gnomon, numerical constrained optimization suggests a Karush–Kuhn–Tucker stationary point at

$$a = 0.2430970\ldots, \quad b = 0.1875777\ldots, \quad c = 0.4515020\ldots, \quad L = 1.2849609182\ldots,$$

at which one self-symmetric edge and one mirror pair of edges of W are tangent to the disk. The rational witness of Theorem 5.1 is a nearby rational point on the feasible side, only 6.16×10^{-7} longer; the exact disk-containment check, not the stationarity computation, is what makes it a theorem. Along the family, the tangency pattern changes twice near $\beta \approx 38.9°$ — the two interior breakpoints of the construction in Section 4.

5. The type example: the golden gnomon, exactly

At a single pleasant triangle the bound becomes a closed-form statement. The *golden gnomon* is the isosceles triangle with unit legs, apex $108°$ (base angles $36°$), and base $\varphi = (1+\sqrt{5})/2$.

Theorem 5.1. *Let T be the golden gnomon, and let S be the polygonal path through*

$$\left(-\frac{2430971}{10^7}, \frac{4515022}{10^7}\right), \quad \left(-\frac{1875779}{10^7}, 0\right), \quad \left(\frac{1875779}{10^7}, 0\right), \quad \left(\frac{2430971}{10^7}, \frac{4515022}{10^7}\right).$$

Then S escapes T, and

$$\ell(T) \leq |S| = \frac{3751558 + 2\sqrt{20693661817348}}{10^7} = 1.28496153349145\ldots < 1.284962.$$

In particular $|S|$ is smaller than Ward's square-path length $1.293473869\ldots$*, by* 8.512×10^{-3}.

Corollary 5.2 (a rigorous bracket)**.** *For the golden gnomon,*

$$\frac{3}{10}\sqrt{25 - 5\sqrt{5}} \;=\; 1.115244103\ldots \;\leq\; \ell(T) \;<\; 1.284962.$$

Figure 7 compares the path of Theorem 5.1 with the four standard candidates at the gnomon, each at a critical placement.

Theorem 5.1 is the case $\beta = 36°$ of Theorem 4.1 with a fixed rational staple, and its certificate is short enough to display in full.

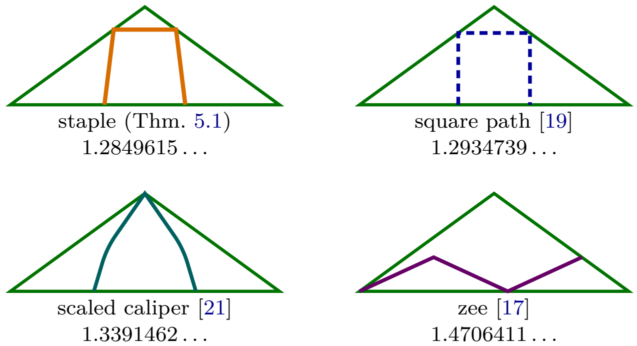

FIGURE 7. The four candidates at the golden gnomon (legs 1, apex 108°, diameter 1.6180340), each drawn at a critical placement: each path's hull fits T only in touching position. The staple, square path, and caliper all touch the base with their endpoints; the caliper's peak sits exactly at the apex. The staple of Theorem 5.1 achieves criticality at the smallest length.

5.1. **Exact data.** Place T with base from $(-\varphi/2, 0)$ to $(\varphi/2, 0)$ and apex $(0, \sin 36°)$; the legs have length 1 since $\cos 36° = \varphi/2$. The sides have lengths $(s_1, s_2, s_3) = (\varphi, 1, 1)$ and unit outward normals

$$n_1 = (0, -1) = u(-90°), \qquad n_{2,3} = (\pm \sin 36°,\ \cos 36°) = u(54°),\ u(126°),$$

and one checks exactly that $\varphi\, n_1 + n_2 + n_3 = 0$. Two golden identities drive all the arithmetic:

$$2\,\mathrm{Area}(T) = \varphi \sin 36° = \sin 72°, \qquad \big(2\,\mathrm{Area}(T)\big)^2 = \frac{5+\sqrt{5}}{8},$$

with $\sin 36° = \frac{1}{4}\sqrt{10 - 2\sqrt{5}}$. All quantities below live in the real field $\mathbb{Q}\big(\sqrt{5}, \sqrt{10-2\sqrt{5}}\,\big)$. The hull H of the staple is the isosceles trapezoid with vertices $(\pm b, 0)$ and $(\pm a, c)$, where

$$a = \frac{2430971}{10^7}, \qquad b = \frac{1875779}{10^7}, \qquad c = \frac{4515022}{10^7},$$

and $|S| = 2b + 2\sqrt{(a-b)^2 + c^2}$ as stated.

5.2. **The Minkowski sum.** By Proposition 2.2 with rotation matrices

$$R_{-\alpha_1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_{-\alpha_{2,3}} = \begin{pmatrix} \pm \sin 36° & \cos 36° \\ -\cos 36° & \pm \sin 36° \end{pmatrix},$$

the path S escapes T iff $W = \varphi R_{-\alpha_1} H \oplus R_{-\alpha_2} H \oplus R_{-\alpha_3} H$ contains $\overline{D}(0, \sin 72°)$. The vertices of W are sums of one vertex from each summand; W is the twelve-edge polygon of Figure 8, symmetric about the horizontal axis (the axial symmetry of H conjugates the two leg copies and fixes the base copy).

5.3. **The twelve inequalities.** For a convex polygon given by a positively oriented vertex cycle $w_1, \ldots, w_{12}$, the inclusion $\overline{D}(0, \rho) \subseteq W$ is equivalent to: (i) all consecutive cross products $(w_{k+1} - w_k) \times (w_{k+2} - w_{k+1}) > 0$ (convexity), (ii) $w_k \times w_{k+1} > 0$ for all k (the origin lies inside), and (iii) for every edge,

$$\big(w_k \times w_{k+1}\big)^2 \ \geq\ \rho^2\, |w_{k+1} - w_k|^2,$$

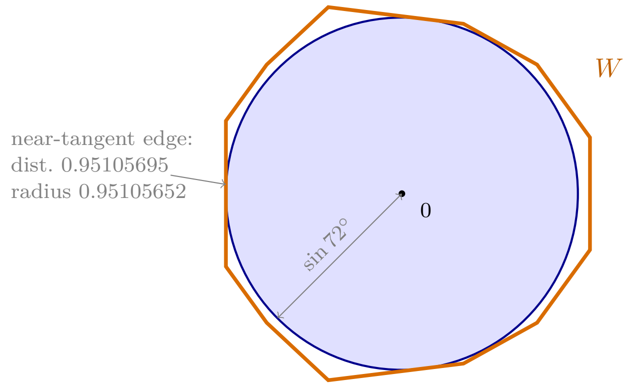

Figure 8. The proof of Theorem 5.1 in one picture: the Minkowski sum $W = \varphi R_{90^\circ} H \oplus R_{-54^\circ} H \oplus R_{-126^\circ} H$ contains the closed disk of radius $2\,\mathrm{Area}(T) = \sin 72^\circ$. Every edge line of W keeps distance $\geq \sin 72^\circ$ from the origin; by Proposition 2.2 the staple escapes. Three of the twelve edges are tangent to the disk to within 5×10^{-7}.

edge of W	multiplicity	distance to 0	certificate D_k
left vertical	1	0.951056952	$+5.131 \times 10^{-7}$
$(-0.951, -0.393) \to (-0.731, -0.697)$	2	1.000620318	$+1.361 \times 10^{-2}$
$(-0.731, -0.697) \to (-0.398, -1.007)$	2	1.007883096	$+2.304 \times 10^{-2}$
$(-0.398, -1.007) \to (0.333, -0.917)$	2	0.951057083	$+5.837 \times 10^{-7}$
$(0.333, -0.917) \to (0.731, -0.697)$	2	0.963597925	$+4.969 \times 10^{-3}$
$(0.731, -0.697) \to (1.016, -0.304)$	2	1.000620318	$+2.287 \times 10^{-2}$
right vertical	1	1.016323686	$+4.731 \times 10^{-2}$
		disk radius $\sin 72^\circ$	0.9510565

Table 1. The distance certificates for the twelve edges of W (mirror-symmetric edges listed once). $D_k > 0$ is the exact inequality (iii); decimals are 60-digit evaluations truncated for display.

i.e., every edge line keeps distance at least ρ from the origin. Here $\rho^2 = \frac{5+\sqrt{5}}{8}$ and all w_k are explicit vectors with entries in $\mathbb{Q}(\sqrt{5}, \sqrt{10 - 2\sqrt{5}}\,)$, so each of (i)–(iii) is the sign of an element of that field. Table 1 lists the distances and the certificate values

$$D_k = (w_k \times w_{k+1})^2 - \rho^2\, |w_{k+1} - w_k|^2;$$

by the axial symmetry of W only seven are distinct. All are strictly positive, with minimum margin $D = 5.13 \times 10^{-7}$.

The ancillary file `tight_staple_proof.py` constructs these field elements symbolically and asks SymPy for their exact algebraic signs; decimal evaluation is used only for the displayed values. It also verifies exactly that all 64 candidate vertex sums lie on or inside every edge half-plane, so the cycle of Table 1 is indeed the boundary of W. Thus the disk is contained in W, and Proposition 2.2 proves that S escapes T.

5.4. **Length comparisons.** With $t = \tan 36° = \sqrt{10 - 2\sqrt{5}}/(1 + \sqrt{5})$:

staple (Thm. 5.1):	$\dfrac{3751558 + 2\sqrt{20693661817348}}{10^7}$	$= 1.2849615\ldots$	
square path [19]:	$\dfrac{3\varphi\, t}{t+2}$	$= 1.2934739\ldots$	$(+8.512 \times 10^{-3})$
scaled caliper [21]:	$\zeta \sin 36°$	$= 1.3391462\ldots$	$(+5.418 \times 10^{-2})$
zee [17]:	$\dfrac{3\varphi\sqrt{5}}{\sqrt{50 + 2\sqrt{5}}}$	$= 1.4706411\ldots$	$(+1.857 \times 10^{-1})$
diameter:	φ	$= 1.6180340\ldots$	$(+3.331 \times 10^{-1})$

All five are explicit algebraic numbers except ζ, for which the directed evaluation $\zeta > 2.2782916$ of its published closed form suffices. This completes the proof of Theorem 5.1. $\square$

Remark 5.3. At $\beta = 36°$ the base-aligned construction gives the valid square path in the table. The leg-aligned value $3s_B = 1.2624233\ldots$ is smaller, but it is not an escape bound there: that square fits strictly inside T (with clearance 6.3×10^{-3}), illustrating why Section 4 uses inscribed squares only as *lower* bounds for escaping square paths.

Proof of Corollary 5.2. Inside T place the isosceles triangle with base angle 45°, with its base on the base of T and its apex at the apex of T. Its height is $h = \sin 36°$ and its legs have length $\sqrt{2}h$. The escape length of a unit-leg 45° triangle is $3/\sqrt{5}$ by [5, 17]. Monotonicity under inclusion and scaling therefore give

$$\ell(T) \geq \sqrt{2}h\,\frac{3}{\sqrt{5}} = \frac{6}{\sqrt{10}}\sin 36° = \frac{3}{10}\sqrt{25 - 5\sqrt{5}}.$$

The upper bound is Theorem 5.1. $\square$

6. Rounding the staple: line–arc paths

The staple is not the last word at the gnomon. Gibbs [12] argued that for a convex polygonal forest some shortest escape path is piecewise linear–circular, the arcs arising as envelopes of placements touching three sides, and his isosceles search [11] proposed line–arc candidates on a range of base angles containing 36°. Free polyline optimization against Proposition 2.2 confirms this at the gnomon: the staple's shoulders relax into shallow circular bends. The second author certified the first sub-staple bound, $\ell(T) \leq 1.2826880$, with a rational 20-gon witness [8]; here we treat the line–arc geometry exactly.

6.1. **An exact line–arc witness.** Proposition 2.2 applies to any compact convex hull, and the window scheme of Remark 2.3 extends as soon as every boundary feature of H has support of the form $\langle p, u\rangle + \varrho\,|u|$ on a known window of directions: vertices ($\varrho = 0$) and circular arcs (p the center, ϱ the radius, on the arc's normal cone). If all feature data are rational, every check is again a finite list of exact signs. Rational line–arc data exist in abundance: on a circle with rational center and rational radius, rational points are dense, the tangent line at a rational point is rational, and a rational chord through a rational point meets the circle in a second one.

Theorem 6.1 (line–arc witness)**.** *There is a symmetric line–arc path S_{arc} (Figure 9) — a bar, two tangent segments, two circular arcs of common rational radius r with rational centers and rational endpoints, and two leg segments, all coordinates rational and listed in the ancillary file* `arc_certificate.py` *— which escapes the golden gnomon, so that*

$$\ell(T) \;\leq\; |S_{\text{arc}}| \;=\; 2q + 2\sqrt{A} + 2\sqrt{B} + 2r\arccos C \;=\; 1.2826798604\ldots \;<\; 1.282680,$$

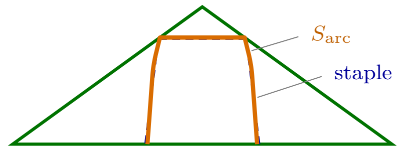

FIGURE 9. The line–arc witness S_{arc} (solid) at its critical placement in the golden gnomon, against the polygonal staple of Theorem 5.1 (dashed). The shoulders bend along circular arcs of radius $r \approx \sin 36°$; the saving is 2.28×10^{-3}.

with $q, A, B, C \in \mathbb{Q}$ explicit.

Proof. The twenty-four rotated feature normals partition the direction circle into eighteen windows. On each window one fixed feature per copy bounds h_W from below by $\langle P, u\rangle + \varrho|u|$; the window minimum is controlled by the two endpoint signs together with an antipode exclusion, and window coverage by the cyclic positivity of consecutive cross products — all exact in $\mathbb{Q}(\sqrt{5}, \sin 36°)$, with minimum squared endpoint slack 1.5×10^{-6}. The single arccos in the length is enclosed by directed evaluation. The ancillary file re-derives all data and performs every check. □

6.2. **The critical line–arc path, exactly.** Fix the template: a bar $[-q, q] \times \{0\}$, corners at $(\pm q, 0)$, tangent edges, two arcs of radius r with normal spans $[\psi_1, \psi_2]$ (and mirror), and legs to the tips. At the numerically optimal member the contact structure is: *tangency along both windows in which the bar-copy arcs face the disk, plus one isolated contact at $u = (-1, 0)$.* Tangency along a window already forces the curvature law: the bar copy carries weight φ, so its arcs map to arcs of radius φr in W, and coincidence with the disk requires $\varphi r = 2\,\mathrm{Area} = \sin 72°$, i.e.

$$r = \sin 36° = \text{width of } T$$

— the same law as the caliper's arcs in the strip, whose radius equals the strip width.

Proposition 6.2 (the critical smoothed staple)**.** *Impose this contact structure and stationarity of the length within the template. After the (linear) elimination of the arc center and leg length, the system reduces to two polynomial equations of bidegree $(4, 4)$ in $t_i = \tan(\psi_i/2)$, with a unique solution in the staple region. Both t_1 and t_2 are algebraic of degree 16 over $\mathbb{Q}$ with palindromic minimal polynomials; e.g. t_1 is a root of*

$$\begin{aligned} 919t^{16} &+ 21152t^{15} + 155604t^{14} + 357968t^{13} - 186516t^{12} \\ &- 1633584t^{11} - 1534804t^{10} + 1227584t^{9} + 3199738t^{8} + 1227584t^{7} \\ &- 1534804t^{6} - 1633584t^{5} - 186516t^{4} + 357968t^{3} + 155604t^{2} \\ &+ 21152t + 919 \;=\; 0. \end{aligned}$$

Consequently $\sin\psi_1, \sin\psi_2$ have degree 8 over $\mathbb{Q}$ and are roots of explicit quartics over $\mathbb{Q}(\sqrt{5})$; the second, for $\sin\psi_2$, reads

$$s^4 + \tfrac{467+103\sqrt{5}}{682}\, s^3 + \left(\tfrac{6\sqrt{5}}{31} - \tfrac{51}{124}\right) s^2 + \tfrac{\sqrt{5}-99}{496}\, s - \tfrac{29+21\sqrt{5}}{5456} \;=\; 0,$$

and the critical length is

$$L^* \;=\; L_{\mathrm{alg}} + 2\sin 36°\,(\psi_2 - \psi_1) \;=\; 1.28267602545904805617809430664645\ldots$$

with L_{alg} algebraic.

The computation is exact throughout: a resultant eliminates t_2; the two quadratic field relations reduce the norm to an integer polynomial of degree 128, whose factorization isolates the degree-16 palindromic factors; the displayed decimal is evaluated from the exact roots (scripts in the repository). Zalgaller's strip constant $\zeta = 2(\tan\alpha + \tilde{\beta} + \tan\gamma)$ is built from angles whose sines satisfy a cubic over $\mathbb{Q}$; the gnomon's critical line–arc constant has the identical shape — algebraic lengths plus a radius times an arc angle entering linearly — one algebraic level up, quartics over $\mathbb{Q}(\sqrt{5})$. That this critical point is the in-template optimum, and that the template is the right structure, rests on the numerics of Sections 4.4 and 7; the *bound* of Theorem 6.1 is unconditional.

6.3. **The upper end of the branch.** The arcs do not persist across the whole staple range. Continuing the line–arc template upward in β, the optimal shoulder-arc span shrinks — 9.5° at 36°, 2.3° at the zee crossing — and vanishes at $\beta \approx 42.9°$ (numerically): above that angle the branch member is exactly polygonal. The polygonal branch then terminates at the right isosceles triangle in a closed-form configuration.

Proposition 6.3 (rectangle endpoint). *For $\beta = 45°$ the rectangle path with vertices $\big(\pm(1-\frac{\sqrt{2}}{2}), 0\big)$ and $\big(\pm(1-\frac{\sqrt{2}}{2}), \sqrt{2}-1\big)$ escapes $T(45°)$, and its length is exactly $\sqrt{2} = 2\cos 45°$, the diameter of $T(45°)$.*

Proof. The hull is the rectangle $H = [-b, b] \times [0, c]$ with $b = 1 - \frac{\sqrt{2}}{2}$ and $c = \sqrt{2} - 1$. The edge normals of the Minkowski sum $W = 2\cos\beta\, R_0(H) \oplus R_1(H) \oplus R_2(H)$ are among the twelve rotated rectangle normals $R_i\{\pm e_1, \pm e_2\}$, and at each of the twelve the exact margin $h_W(\nu) - \rho$, computed in $\mathbb{Q}(\sqrt{2})$, equals either 0 (ten directions) or $3\sqrt{2}/2 - 2 > 0$ (two directions). Hence $W \supseteq \rho D$ and Proposition 2.2 applies; the length is $2b + 2c = \sqrt{2}$. □

Ten active constraints out of twelve is a maximally degenerate configuration, as one expects at the end of a branch: numerically, for β just above 45° the trapezoid family collapses onto the diameter segment, whose length the endpoint already matches.

7. The caliper regime and the Zalgalloid junction

Everything in this section is numerical; it enters no proof. Below the staple range the width mechanism of the introduction takes over: the scaled caliper — Zalgaller's strip solution scaled to the width $\sin\beta$ of $T(\beta)$ — escapes at length $\zeta \sin\beta$, and below $\beta^* \approx 27.6°$ nothing known beats it. What remains is to understand the handover, and Ward guessed its shape in one word: he asked whether the class optimal beyond the zee range might be a family of "Zalgalloids" containing the caliper as an instance. Numerically the answer is yes, the smoothed staple is its upper end, and the handover is tangential.

Enlarge the template of Section 6.2 by an optional foot arc (radius again $\sin\beta$) at each bar corner; a member is then a convex line–arc hull, and the path is its boundary minus the chord joining the two tips. Continuation in β gives Figures 10 and 11: as β decreases from 36° the bar shrinks and the arcs deepen; the family beats the scaled caliper strictly for all β down to $\approx 27.75°$ — well below the polygonal staple's caliper crossing at 31.05° — with margin collapsing from 5.3×10^{-3} at 31° to 1.0×10^{-5} at 27.75°; between 27.5° and 27.75° the improvement vanishes, and by 27.25° the optimized member *is* the scaled caliper to solver precision: the bar length reaches zero (the vanished bar is the caliper's peak corner), the arc span locks onto Zalgaller's $\tilde{\beta} = 18.27°$, and the tips become the caliper's feet. The family thus interpolates continuously from the trapezoid to the caliper itself — precisely the family Ward speculated about, with the polygonal staple at one end and Zalgaller's caliper at the

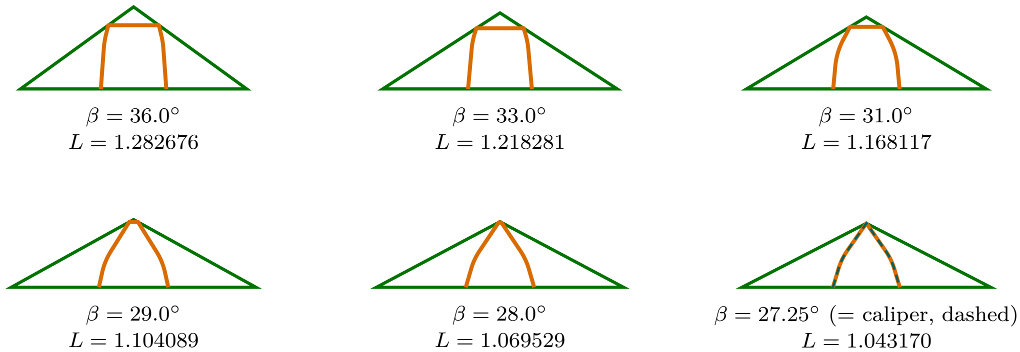

FIGURE 10. The Zalgalloid family (numerical): optimized line–arc members at six base angles, each at its critical placement (tips on the base, bar or peak under the apex). As β decreases the bar shrinks and the arcs deepen; at $\beta = 27.25°$ the optimized member coincides with the scaled Zalgaller caliper (dashed): the vanished bar has become the caliper's peak corner, the shoulder arcs its arcs (span $\tilde{\beta} = 18.27°$), the tips its feet.

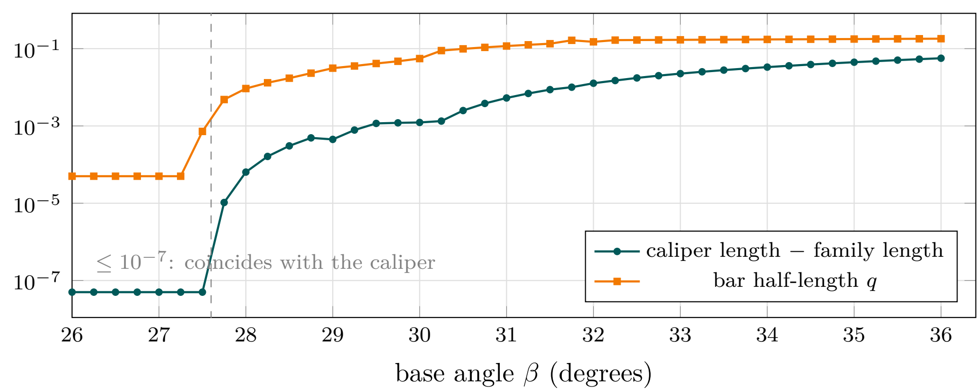

FIGURE 11. Convergence of the line–arc family onto the scaled caliper (numerical; log scale). Both the improvement over the caliper and the bar length vanish at $\beta^* \approx 27.6°$; clipped values at the floor of the plot are exact coincidence to solver precision.

other. Making the junction exact is an inviting next target. Together with Proposition 6.3, the picture is of one connected boundary-type branch running from the exact rectangle at $\beta = 45°$, through the polygonal trapezoids ($\beta \gtrsim 42.9°$) and the smoothed staples, to Zalgaller's caliper at β^*; its reign as the best-known frontier is precisely $[\beta^*, \beta_\times]$.

The line–arc "tunnel" candidates Gibbs sighted below his trapezium phase [11] are, we believe, members of this family.

8. Conclusion

The picture this paper leaves is a phase diagram (Figure 1). Across $26° \leq \beta \leq 60°$ the best-known escape paths form three regimes — caliper, the staple-to-Zalgalloid branch, zee — separated by a second-order boundary at $\beta^* \approx 27.6°$, where the branch merges tangentially into the caliper, and a first-order boundary at $\beta_\times = 42.287°$, where two combinatorially distinct families cross transversally and the upper-bound envelope attains its maximum: the hardest isosceles forest sits at the phase boundary. The exact contributions are the anchors inside the middle regime: a certified upper bound on the whole interval $[31.07°, 42.16°]$ of base angles, via an explicit trapezoidal family that removes Ward's square path from the frontier everywhere and beats all four of his comparison families on $[31.50°, 42.16°]$; a finite-certificate method (for polygonal and line–arc data alike) that replaces the continuum of placements by exact arithmetic; the line–arc bound $\ell(T) \leq 1.2826799$ at the golden gnomon; the algebraic characterization of the critical line–arc path there, with its Zalgaller-form constant; and the exact rectangle endpoint of the branch at $\beta = 45°$ (Proposition 6.3). The numerical contributions — the curvature law $r = \sin\beta$, the location and transversality of $\beta_\times$, the tangential junction at β^* — mark the places where theorems should come next: a proof of the curvature law via the linear-programming duality it suggests, certified interval bounds for the line–arc family (closing in particular the uncertified sliver $(42.16°, \beta_\times)$), and an exact treatment of both boundaries. No optimality is claimed below 45°; the gap in Corollary 5.2 remains substantial.

Ancillary files.

- `tight_staple_proof.py` verifies Theorem 5.1: it reconstructs W, determines the exact algebraic signs in Table 1, and checks the length comparisons.
- `interval_construction.json` lists the exact rational coefficients of the staple family $S(u)$.
- `interval_certificate.py` verifies all 116 Sturm certificates of Theorem 4.1 and Corollary 4.2.
- `arc_certificate.py` constructs the rational line–arc witness of Theorem 6.1 and verifies its eighteen window certificates and the length enclosure.

All four files, together with the numerical discovery code and its data, are maintained at `https://github.com/atemerev/bellman-staple`.

Appendix A. The coefficients of the staple family

Table 2 prints the construction of Theorem 4.1 in full. The same numbers are listed as integer fractions in the ancillary file `interval_construction.json`, together with the fixed support-test vertex choices used in the escape certificates; the verification script consumes the file, not this table.

piece 1: $u \in [139/500,\ 353/1000]$			
k	a_k	b_k	c_k
0	0.2180623261593	0.50834630972286	−0.68618491542384
1	−2.98683316746864	−9.67586466442302	16.00857768415038
2	34.80963432476292	88.60308231720582	−120.98302430509386
3	−165.44093496652428	−407.4340115522607	544.85319850303314
4	419.36689348412628	1052.77262491939776	−1412.41888562475624
5	−566.71412345478798	−1463.96980815258684	1965.72502529223156
6	317.71808396659464	860.8859714000805	−1153.82147540598876
piece 2: $u \in [353/1000,\ 353399/1000000]$			
k	a_k	b_k	c_k
0	−44.64121070435796	64.82150688020952	−53.83538739921426
1	258.38271230989518	−370.2997274914329	310.226332252557
2	−371.75125098832224	530.4122974737888	−443.04905105780508
piece 3: $u \in [353399/1000000,\ 771/2000]$			
k	a_k	b_k	c_k
0	598.39329462654	931.9350917140677	−1000.1742396697317
1	−9888.67910670574158	−15411.81001925178432	16547.06303226795252
2	68135.45237045777328	106266.35727563781996	−114104.67388908582498
3	−250502.62369568886828	−391014.4027686377397	419956.42011173402622
4	518368.51791825810486	809875.46729334615888	−870074.61994937309994
5	−572502.14475386179926	−895368.13774997390718	962241.70033218066126
6	263679.85825187420358	412866.98586667984212	−443874.26394488674308

Table 2. The 51 coefficients of the staple family: on each piece, $x(u) = \sum_k x_k u^k$ for $x \in \{a, b, c\}$. Every value is an exact rational number whose denominator has the form $2^i 5^j$, printed here as its full terminating decimal.

References

[1] A. Adhikari and J. Pitman, *The shortest planar arc of width 1*, Amer. Math. Monthly **96** (1989), 309–327.
[2] R. Bellman, *Minimization problem*, Bull. Amer. Math. Soc. **62** (1956), 270.
[3] A. S. Besicovitch, *On arcs that cannot be covered by an open equilateral triangle of side 1*, Math. Gazette **49** (1965), 286–288.
[4] T. M. Chan, A. Golynski, A. López-Ortiz and C.-G. Quimper, *Curves of width one and the river shore problem*, Proc. 15th Canad. Conf. Comput. Geom. (CCCG), 2003, 73–75.
[5] P. Coulton and Y. Movshovich, *Besicovitch triangles cover unit arcs*, Geom. Dedicata **123** (2006), 79–88.
[6] Z. Deng, *A general solution to Bellman's lost-in-a-forest problem*, arXiv:2412.10686v3 (2025), https://arxiv.org/abs/2412.10686v3.
[7] Z. Deng, *Proof and more variations of Bellman's lost-in-a-forest problem*, arXiv:2606.13987v2 (2026), https://arxiv.org/abs/2606.13987.
[8] A. Doria, *Smoothed-Staple: certified escape paths beyond the staple*, code repository (2026), https://github.com/Alessio2405/Smoothed-Staple.
[9] S. R. Finch and J. E. Wetzel, *Lost in a forest*, Amer. Math. Monthly **111** (2004), 645–654.
[10] J. Gerriets and G. Poole, *Convex regions which cover arcs of constant length*, Amer. Math. Monthly **81** (1974), 36–41.
[11] P. Gibbs, *Lost in an isosceles triangle*, viXra:1607.0015 (2016).
[12] P. Gibbs, *Bellman's escape problem for convex polygons*, viXra:1606.0050 (2016).
[13] O. Gross, *A search problem due to Bellman*, RAND Corporation, Research Memorandum RM-1603, 1955.
[14] J. R. Isbell, *An optimal search pattern*, Naval Res. Logist. Quart. **4** (1957), 357–359.
[15] H. Joris, *Le chasseur perdu dans la forêt*, Elem. Math. **35** (1980), 1–14.
[16] E. Lutwak, *Containment and circumscribing simplices*, Discrete Comput. Geom. **19** (1998), 229–235.
[17] Y. Movshovich, *Besicovitch triangles extended*, Geom. Dedicata **159** (2012), 99–107.
[18] Y. Movshovich and J. E. Wetzel, *Escape paths of Besicovitch triangles*, J. Combinatorics **2** (2011), 413–433.
[19] J. W. Ward, *Exploring the Bellman forest problem*, manuscript (2008), http://wardsattic.com/math/BellmanForestProblem/BellmanForestProblem.pdf; archived at https://web.archive.org/web/20250216114838/http://wardsattic.com/math/BellmanForestProblem/BellmanForestProblem.pdf.
[20] W. Weil, *Decomposition of convex bodies*, Mathematika **21** (1974), 19–25.
[21] V. A. Zalgaller, *How to get out of the woods? On a problem of Bellman* (Russian), Matematicheskoe Prosveshchenie **6** (1961), 191–195.

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